



# Service Tool Instruction

Service Tool Instruction  
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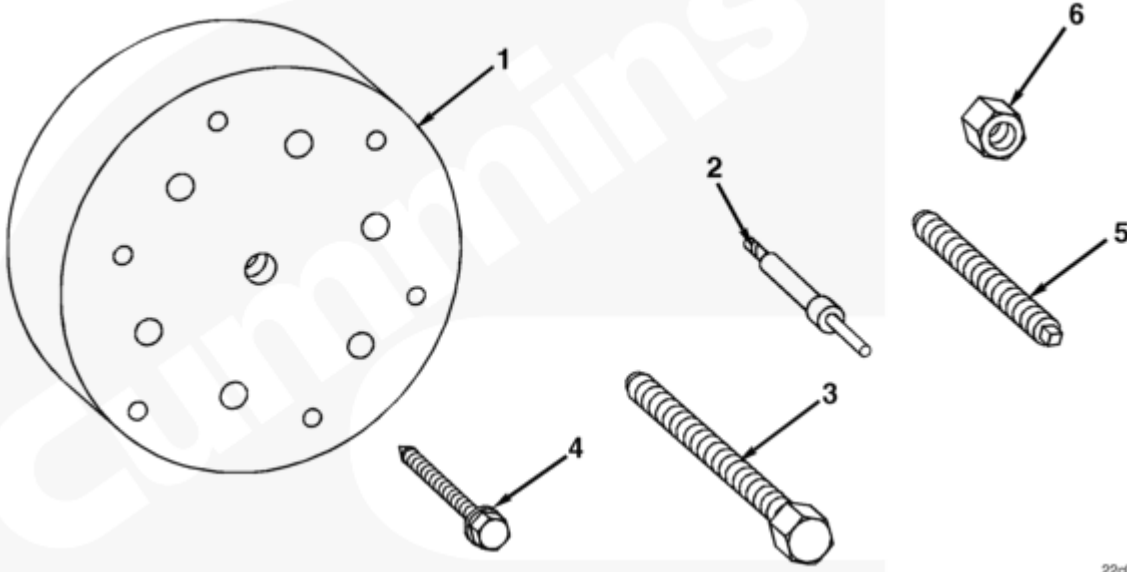
## Description

Rear Crankshaft Seal Replacer Kit

## Purpose

This service tool instruction provides information for the use of the Signature, ISX, and QSX15 rear crankshaft seal replacer kit, Part Number 3164780. The rear crankshaft seal replacer kit, Part Number 3164780, has also been designed for use on the QSX15 rear power takeoff (REPTO) engine. \*

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**Table 1, Rear Crankshaft Seal Replacer Kit, Part Number 3164780**

| Item | Part Number | Description              | Quantity    |
|------|-------------|--------------------------|-------------|
| 1    | 3164763     | Replacer plate           | 1           |
| 2    | 3163089     | Pilot drill              | 1           |
| 3    | 3164154     | Replacer screw           | 1           |
| 4    | 3164781     | Self-tapping metal screw | 10          |
| 5    | 3164155     | Replacer screw           | 3 (1 shown) |
| 6    | 3164156     | Regular hexagon nut      | 3 (1 shown) |

**Table 2, Items Used with Rear Crankshaft Seal Replacer Kit (REPTO Applications Only), Purchased Separately**

| Item      | Part Number | Description  | Quantity |
|-----------|-------------|--------------|----------|
| Not shown | 3164765     | Guide pin ** | 3        |

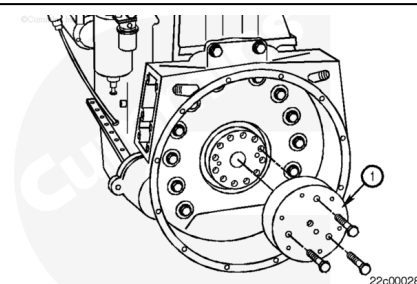
\* The existing replacer plate, Part Number 3162995, included in the rear crankshaft seal kits, Part Number 3164153 and Part Number 3162994, can be modified by drilling three additional 17-mm [21/32-inch] holes. The six 17-mm [21/32-inch] holes are equally spaced on a 138-mm [5.433-in] bolt circle. The additional three holes update rear crankshaft seal replacer, Part Number 3162995, to the rear crankshaft seal replacer, Part Number 3164763.

\*\* Included in the REPTO Replacer Kit, Part Number 3164761.

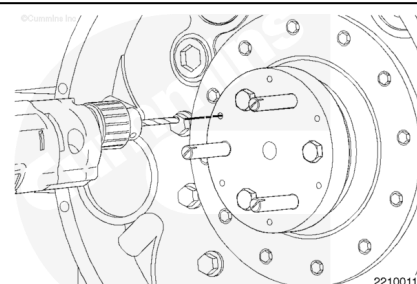
**Note :** Replacer screw, Part Number 3164155 and hexagon nut, Part Number 3164156 are required on engines with flywheel mounting capscrews that are too short.

Install the replacer plate (1), Part Number 3164763, onto the crankshaft using three of the flywheel mounting capscrews or three replacer screws and three hexagon nuts.

Tighten the capscrews or hexagon nuts finger-tight until the replacer plate is positioned against the seal.

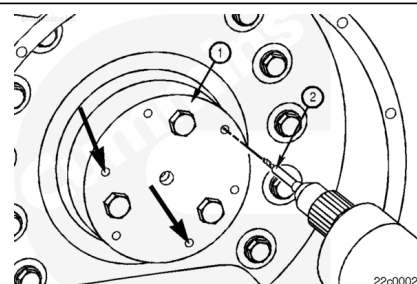


Three guide pins, Part Number 3164765, are required on the REPTO application to support the REPTO crankshaft gear and the replacer plate.



**Note :** Although six holes are provided in the replacer plate (1), a minimum of three are required to remove the crankshaft seal. If there is difficulty in removing the seal, all six holes can be used.

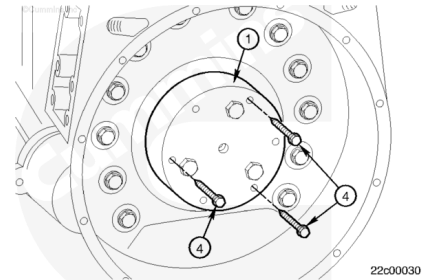
**Note :** Applying grease on the pilot drill (2), Part Number 3163089 will help catch the metal chips and prevent them from entering the oil pan.



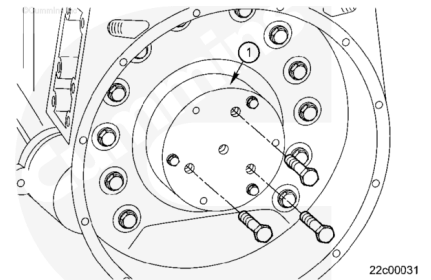
Place the pilot drill (2) into one of the six holes in the replacer plate (1). Drill a hole through the crankshaft seal casing.

Remove the pilot drill (2) from the replacer plate (1), and repeat the above procedure using every other hole in the replacer plate (1).

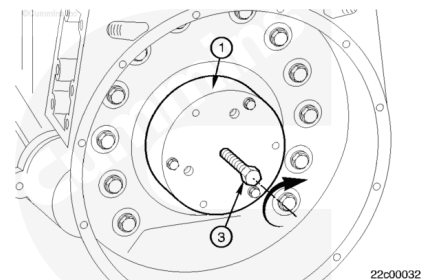
Install the self-tapping metal screw (4), Part Number 3164781 through the replacer plate (1) into each of the drilled holes in the crankshaft seal.



Remove the three flywheel mounting capscrews, or the replacer screws and hexagon nuts from the replacer plate (1).



Install the replacer screw (3), Part Number 3164154 into the center hole of the replacer plate (1), and turn **clockwise** until the crankshaft seal is removed from the crankshaft.

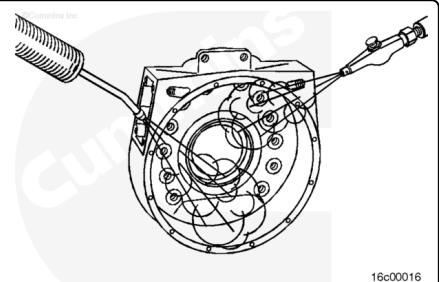


Clean the crankshaft seal mounting surface of the flywheel housing.

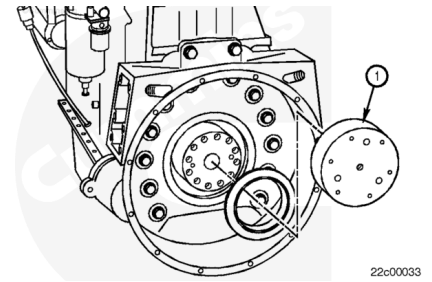
Use a crocus cloth to remove any rust or other deposits from the crankshaft sealing surface.

Use a lint-free cloth to remove any traces of oil from the crankshaft sealing surface.

Inspect the crankshaft sealing surface for nicks, burrs, and grooves. Damage to the sealing surface can result in seal leakage.



**Do not use a hammer or mallet to install the seal. Doing so can damage the seal and/or crankshaft.**

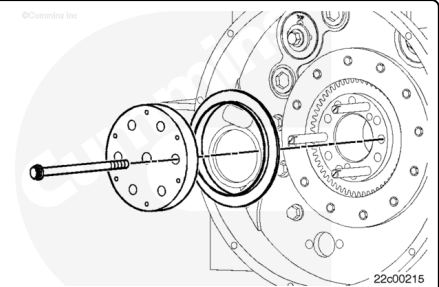


**Note :** A solution with 10 parts of water and one part soap must be used around the outer rubber surface of the crankshaft seal to prevent binding and improve bonding. No other lubricants are to be used during the installation process.

Position the crankshaft seal and replacer plate (1), Part Number 3164763, over the end of the crankshaft.

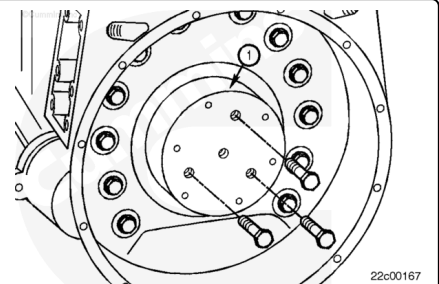
On REPTO applications, three guide pins, Part Number 3164765, are used to support the REPTO crankshaft gear and the replacer plate.

Position the crankshaft seal and replacer plate, Part Number 3164763, over the guide pins and REPTO crankshaft gear.



Install three of the flywheel mounting capscrews, or three replacer screws, Part Number 3164155 and three hexagon nuts, Part Number 3164156 through the replacer plate (1) and into the end of the crankshaft.

Tighten the capscrews or the hexagon nuts finger-tight.



**⚠ CAUTION ⚠**

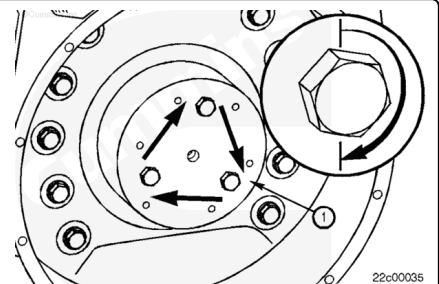
**The crankshaft seal must be pressed evenly into the bore area. If the seal is not pressed into the bore area evenly, damage to the seal can result.**

Tighten the capscrews or the hexagon nuts no more than 1/2 of a revolution at a time, alternating capscrews in a **clockwise** pattern until the replacer plate (1) is positioned against the flywheel housing.

**Torque Value:** 16 n·m [ 142 in-lb ]

Remove the replacer plate (1) from the crankshaft.

Inspect the crankshaft seal position to make certain the seal face is positioned below the flywheel housing bore taper and squarely positioned in the bore.



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**Last Modified: 16-Jan-2003**

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